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(54) **MULTIPURPOSE DEVICE WITH CONTROLLABLE PHOTONIC AND ELECTROMAGNETIC MODALITIES**

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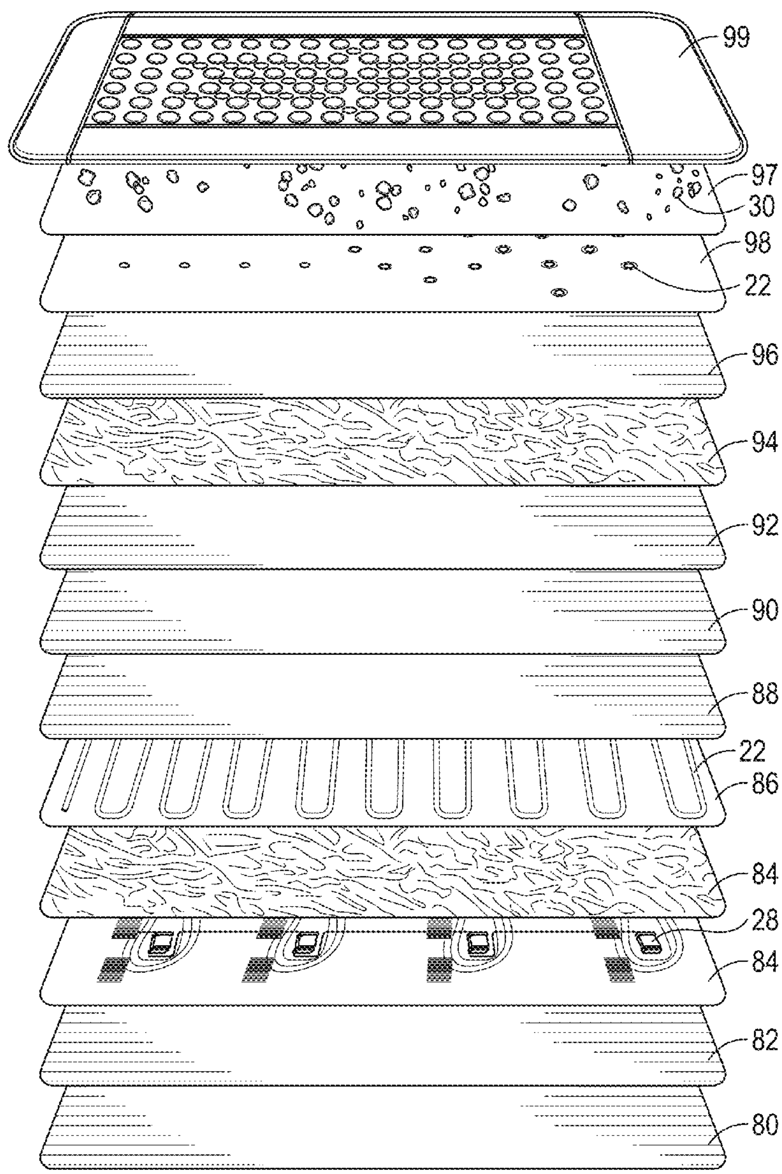
(57) **ABSTRACT**

A device for providing controlled photonic modalities includes a substrate having a length and a width, at least one array of controllable photonic emitters and at least one electromagnetic emitter both attached the substrate. A controller is configured to allow a user to select the type of photonic energy and electromagnetic energy.



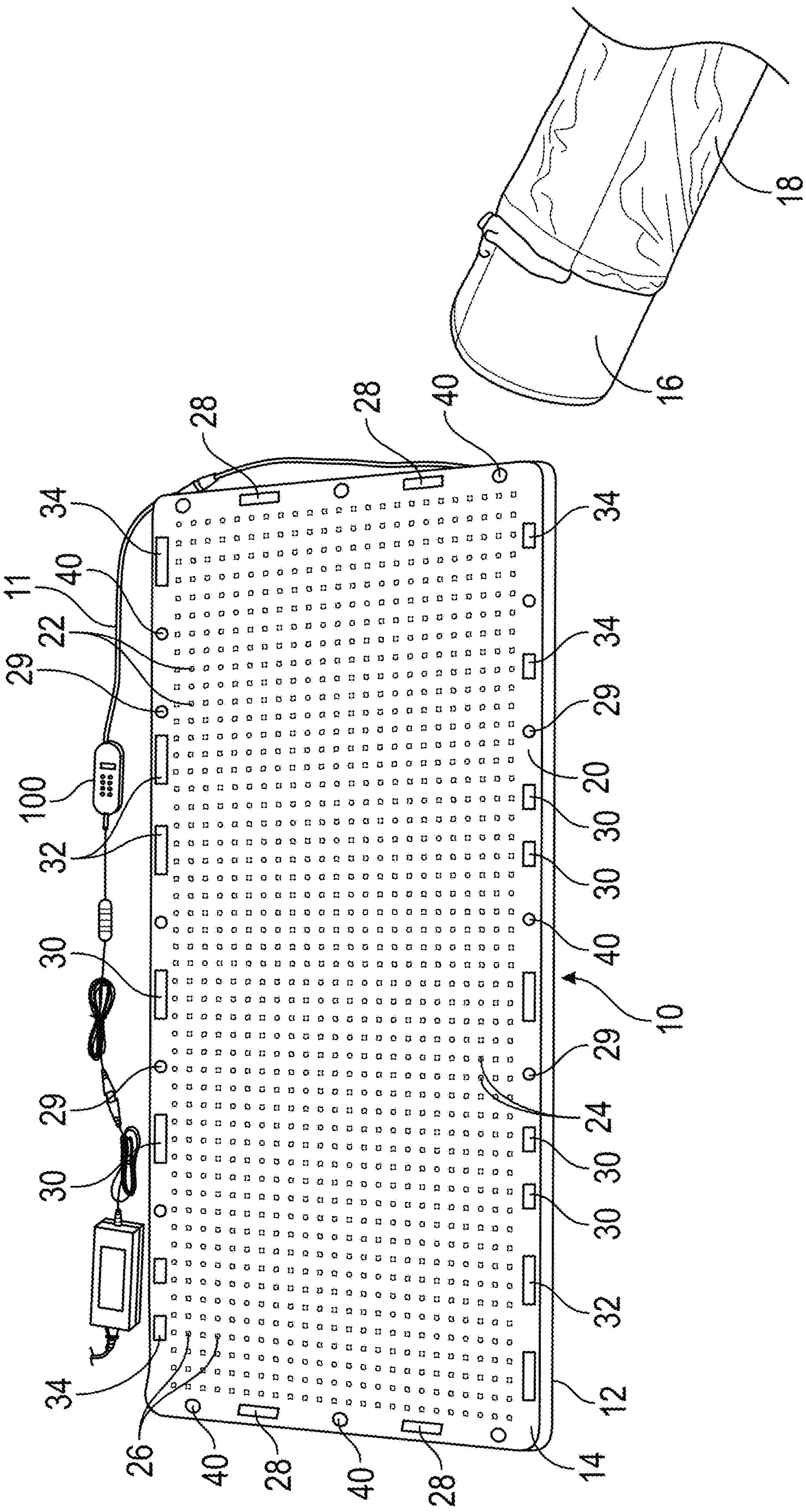

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FIG. 1

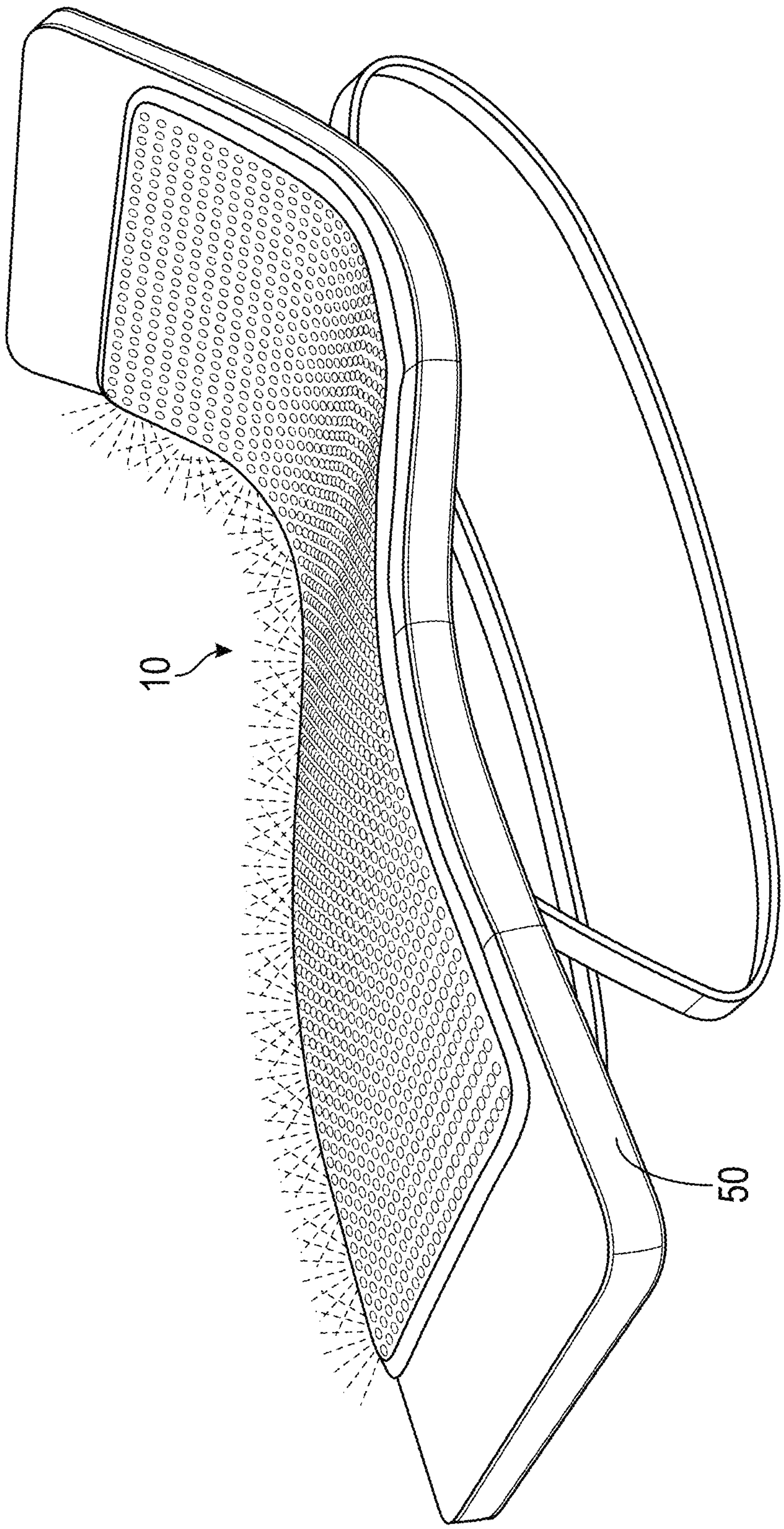


FIG. 2

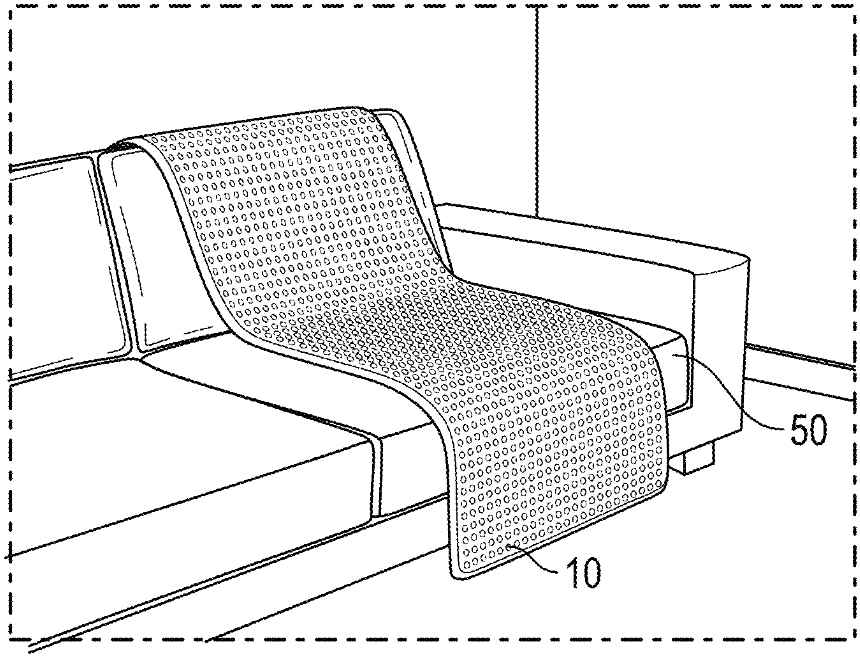


FIG. 3

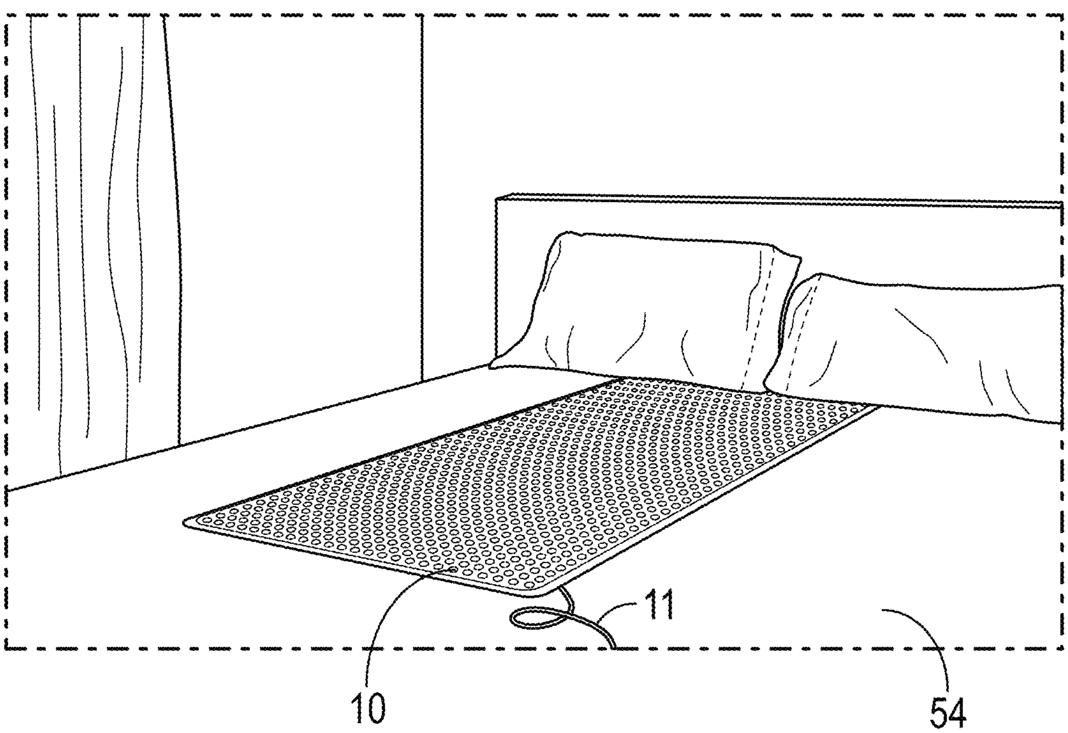


FIG. 4

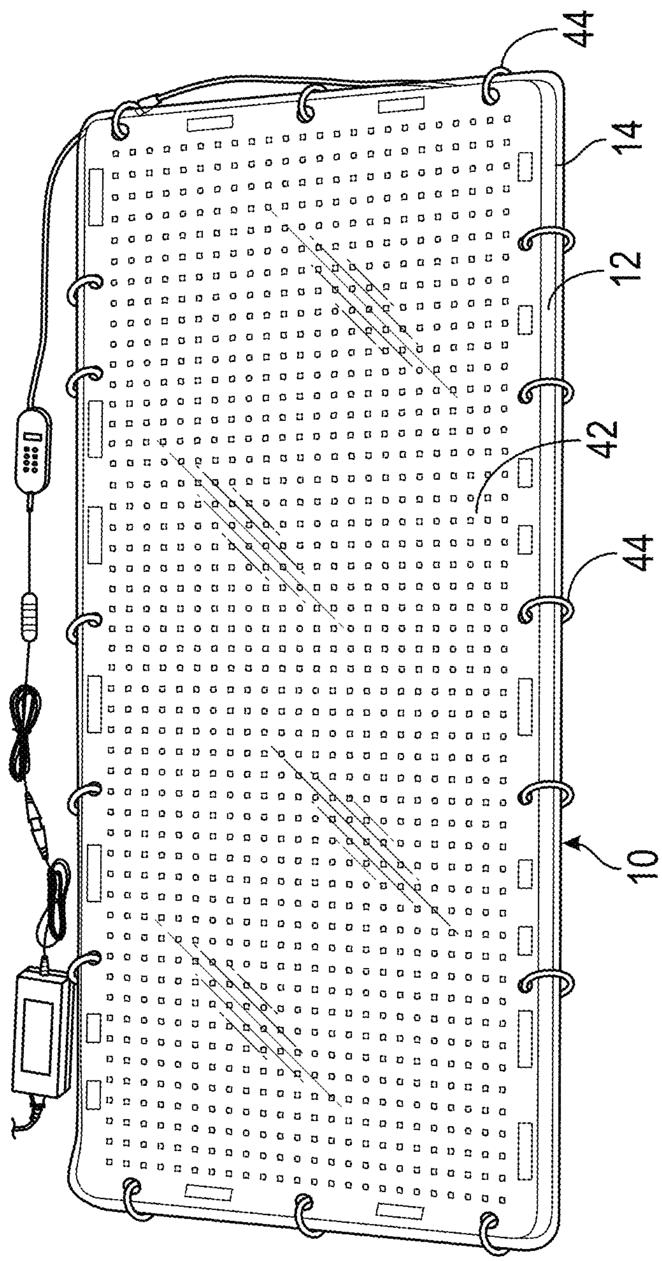


FIG. 5

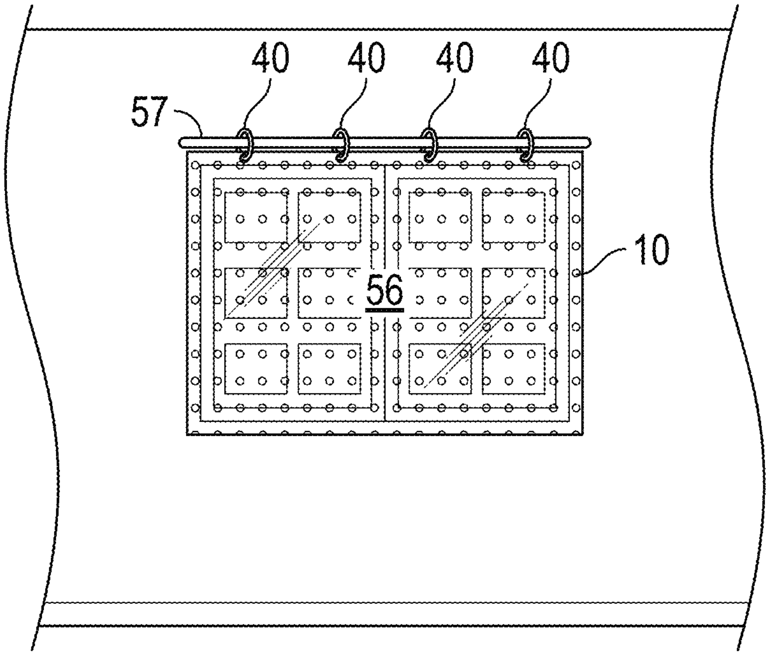


FIG. 6

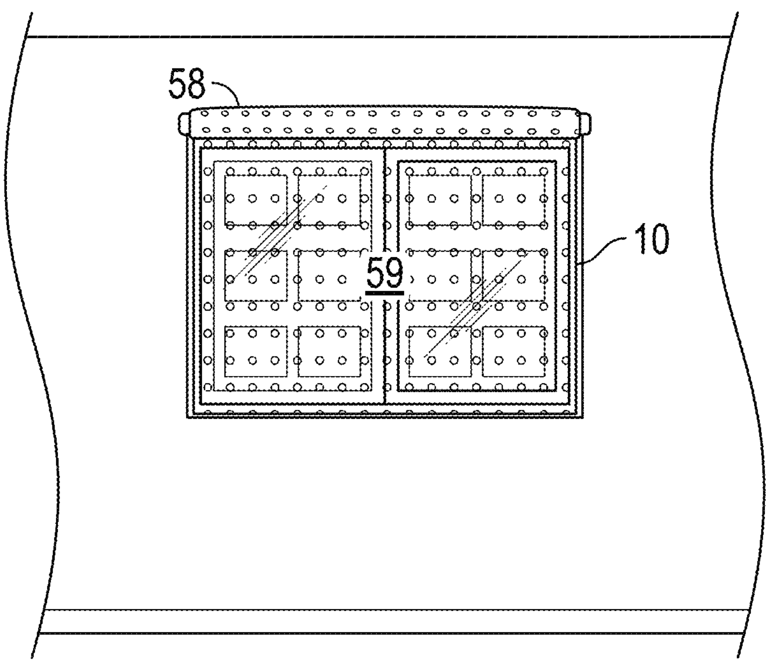


FIG. 7

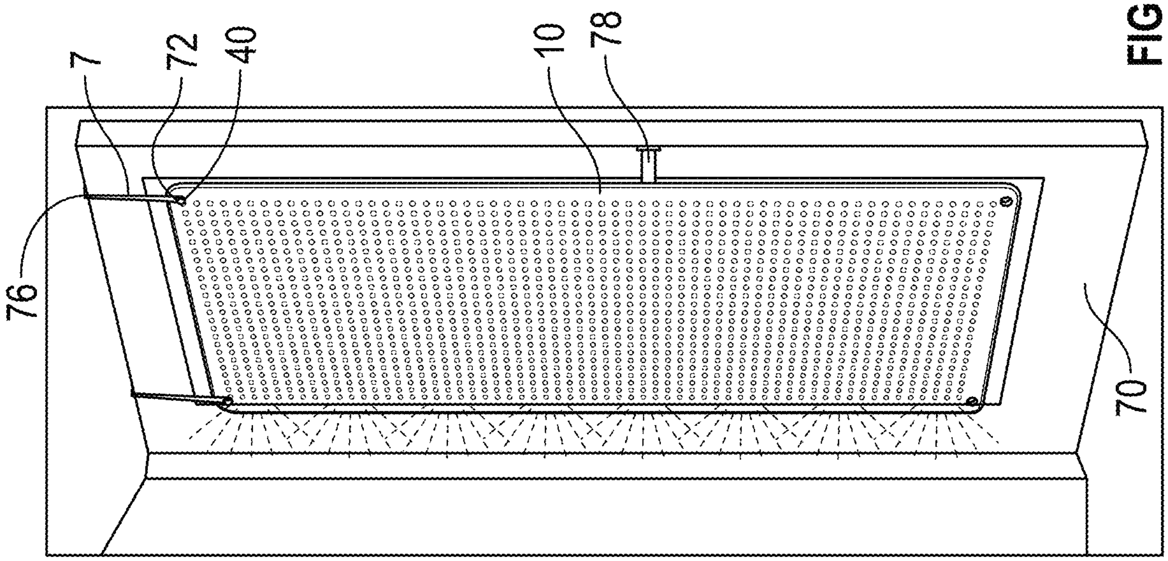


FIG. 9

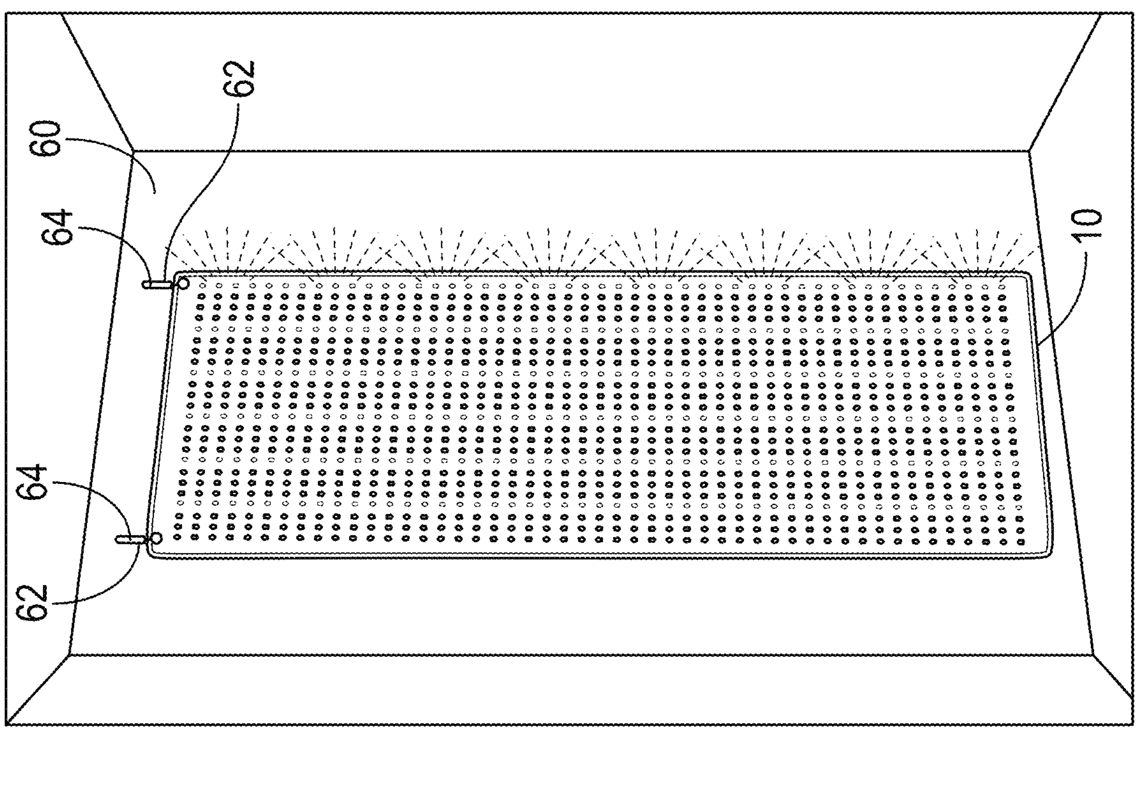


FIG. 8

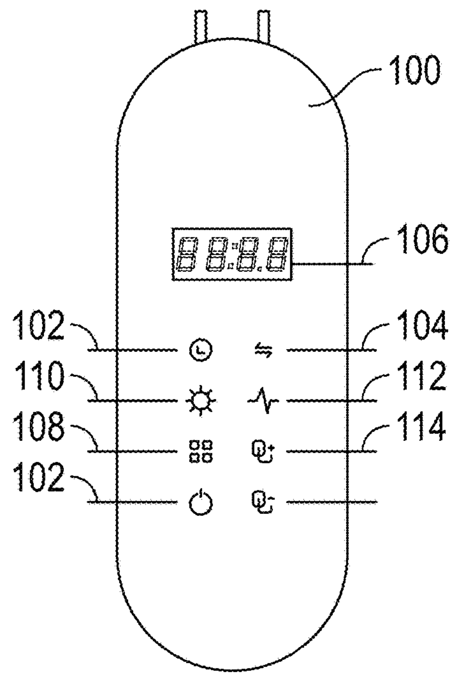
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FIG. 10

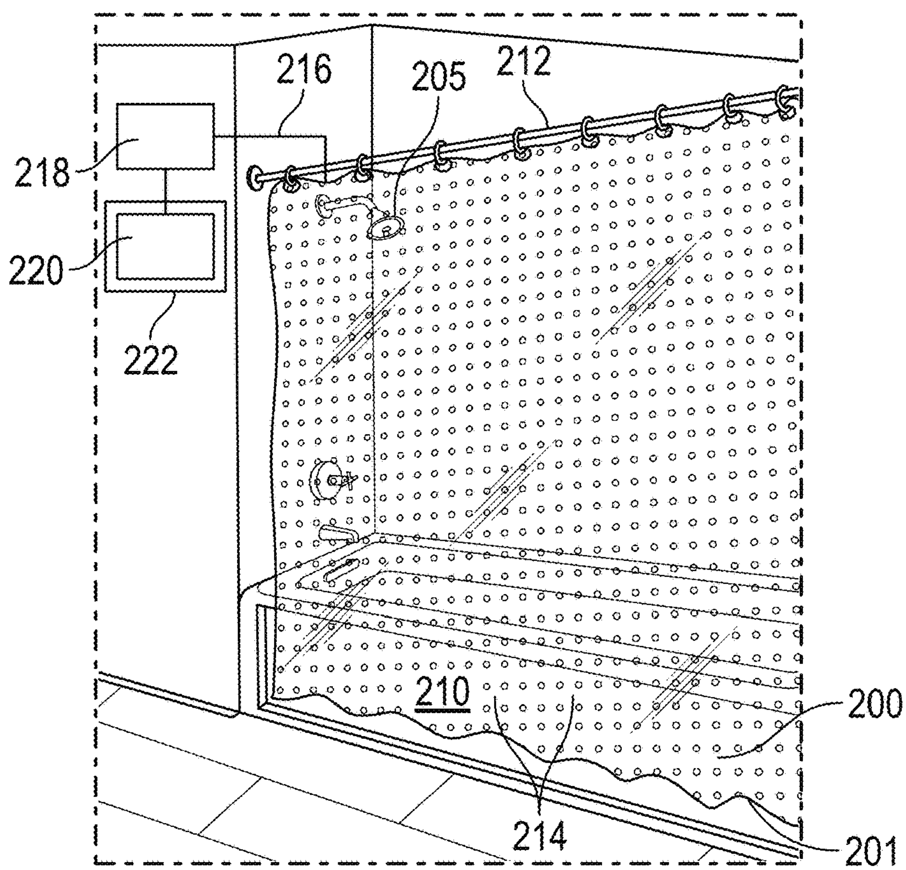
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FIG. 11

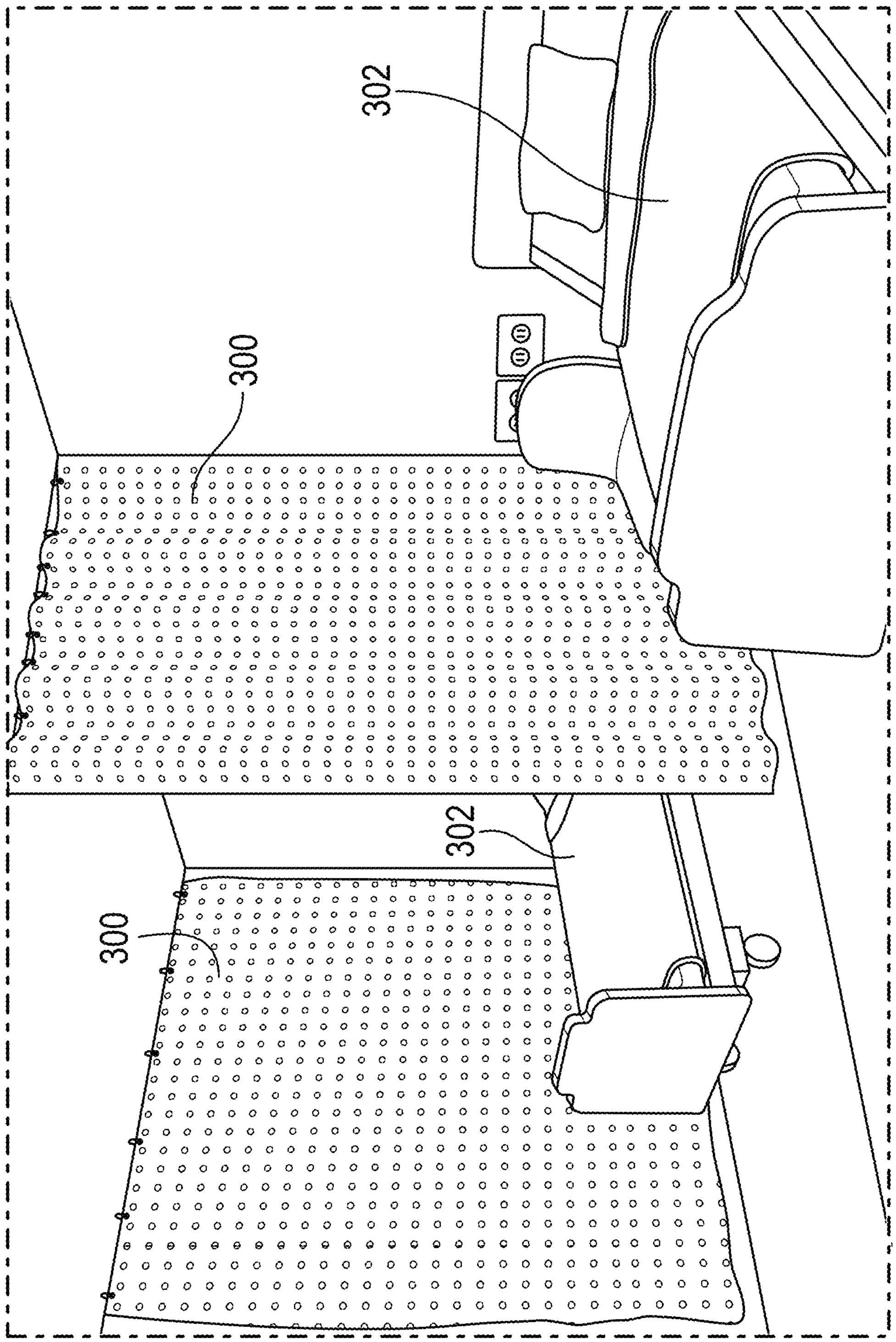


FIG. 12

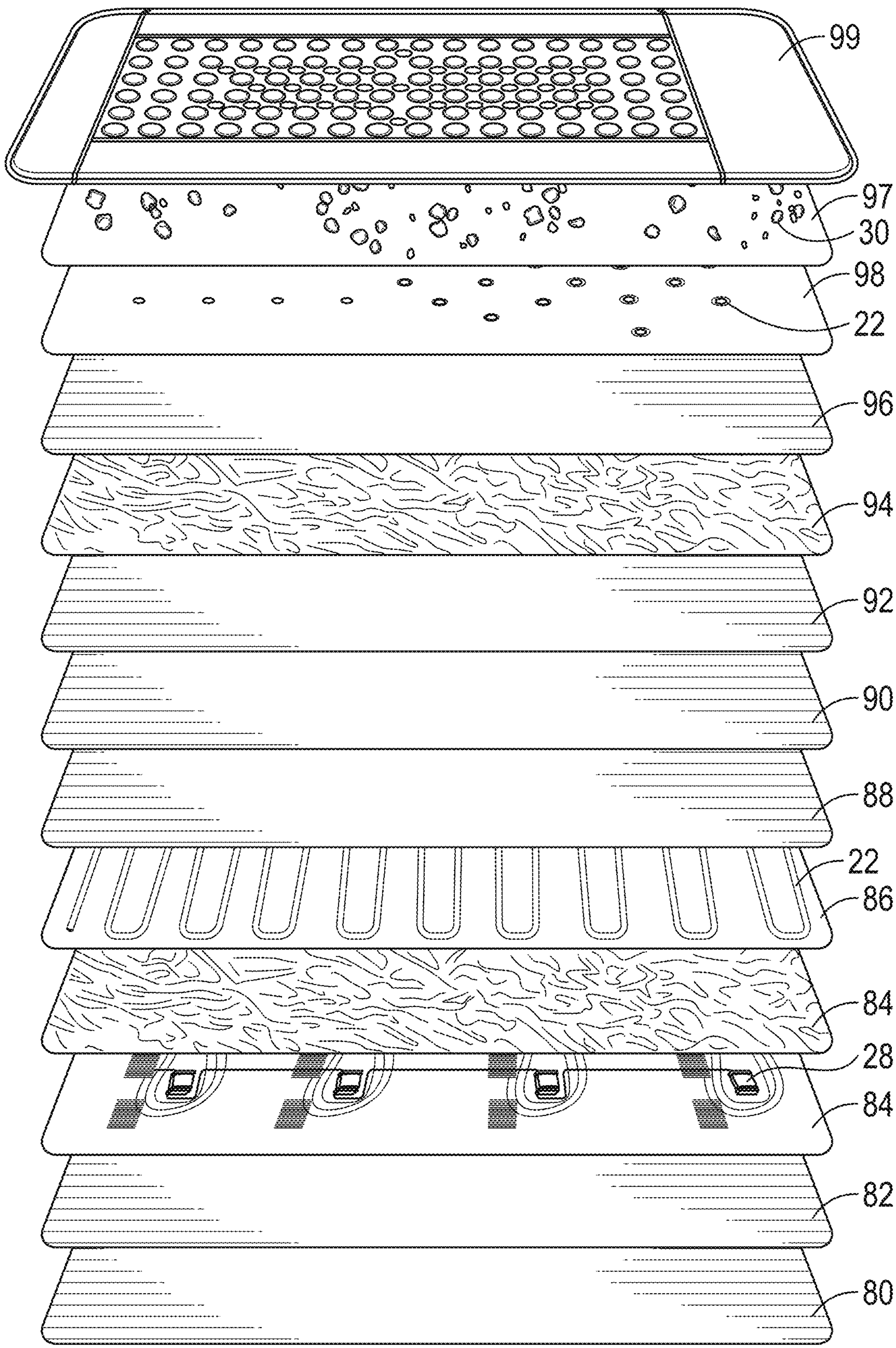


FIG. 13

# MULTIPURPOSE DEVICE WITH CONTROLLABLE PHOTONIC AND ELECTROMAGNETIC MODALITIES

## CROSS REFERENCE TO RELATED APPLICATION(S)

**[0001]** The present application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/561,470 entitled MULTIPURPOSE DEVICE WITH CONTROLLABLE PHOTONIC AND ELECTROMAGNETIC MODALITIES that was filed on Mar. 5, 2024, the contents of which are incorporated by reference in its entirety.

## BACKGROUND

**[0002]** The present disclosure relates a multipurpose device that is configured for use as a mat, a cover for draping on a surface or as a hanging sheet. More particularly, the present disclosure relates to a multipurpose device having controllable photonic and/or electromagnetic modalities that can provide one or more treatments to a person, provide ambiance to a space and/or change environmental conditions within a space.

**[0003]** Many people desire to utilize photonic energy for treatments or to enhance the effects of exercise, physical therapy, massage and the like. However, to obtain the photonic treatments, the user typically has to travel to a location, which can be inconvenient.

**[0004]** Additionally, even if a user were able to obtain a device for the photonic treatment, the devices can be bulky and difficult to store. Especially where a user's living space is limited, the size of the photonic device may prevent a user from obtaining the photonic device.

**[0005]** The different photonic energies provide different benefits to the user. By way of non-limiting example, red light (620-750 nm) in the visible light spectrum is known to improve skin appearance, such as reducing wrinkles, scars, redness and acne and aid in recovery after working out or experiencing trauma to the person's body. Green light (525-550 nm) is known to improve the appearance of the skin by lessening dark circles, pigmentation, broken capillaries and sunspots and calm irritated or over-stimulated skin. Blue light therapy (405-410 nm) can be used to treat acne and can improve skin tone and texture. Yellow light therapy (580-590 nm) can reduce the appearance of solar lentigos. Full spectrum white light (400-700 nm) can uplift mood and enhance meditative calm.

**[0006]** The photonic energies also include those in the infrared spectrum (800-1200 nm) which can be used for detoxification, pain relief, reduction of muscle tension, relaxation, improved circulation, weight loss, skin purification, boosting of the immune system, aid in lowering of blood pressure and aid in recovery from physical exercise. The photonic energies also include those in the ultraviolet spectrum (280-380 nm) which can be used to treat skin disorders such as psoriasis, vitiligo, destroy bacteria and promote wound healing.

**[0007]** The use of electromagnetic modalities creates magnetic fields that can improve the human body's natural recovery and energy, and provide treatments or biomagnetic therapies for users. The electromagnetic modalities can be used in physical recovery, healing, to lessen pain, management of diabetes, recovery from injury, assist musculoskeletal conditions, tissue repair or inflammation, or conditions including bone fractures and degenerative conditions. Additionally, electromagnetic treatments can aid in adjusting the pH in a user's body to reestablish a balanced and substantially neutral pH.

**[0008]** There is a need to deliver photonic and/or electromagnetic modalities in a controlled manner with a device that is configured to cover a large and/or selected area for a desired treatment while being able to be rolled and/or folded for storage in a compact configuration.

## SUMMARY

**[0009]** One aspect of the present disclosure includes a device for providing controlled photonic modalities and pulsed electromagnetic frequency modalities. The device includes a substrate having a length and a width, at least one array of controllable photonic emitters attached the substrate and an electromagnetic emitter. A controller is configured to allow a user to select the type of photonic energy and electromagnetic energy.

**[0010]** Another aspect of the present disclosure relates to a device for providing controlled photonic modalities and pulsed electromagnetic frequency modalities. The device includes a base layer having an upper surface and a lower surface configured to engage a surface and a first internal layer supporting pulsed electromagnetic field motor, the first internal layer having a lower surface facing the upper surface of the base layer and an upper surface. The device includes a second internal layer supporting one or more infrared emitters, wherein the second internal layer having a lower surface facing the first internal layer and an upper surface and a first internal insulating layer between the first internal layer and the second internal layer. The device includes a third internal layer supporting a plurality of visible light emitters, the third internal layer having a lower surface facing the upper surface of the second internal layer and a second internal insulating layer between the second internal layer and the third internal layer. The device includes a fourth internal layer supporting a plurality of stone that emit photons when heated, the fourth internal layer having a lower surface facing the upper surface of the third internal layer and an upper surface and an upper layer having a substantially optically transparent area, the upper layer having a lower surface facing the upper surface of the fourth internal layer and an upper surface configured to engaged by a user.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0011]** FIG. **1** is a schematic view of a multipurpose device with controlled photonic and/or electromagnetic modalities in a stored configuration and a configuration for use.

**[0012]** FIG. **2** illustrates the device on a lounge chair.

**[0013]** FIG. **3** illustrated the device on a couch.

**[0014]** FIG. **4** illustrates the device on a bed.

**[0015]** FIG. **5** illustrates the device with a plastic covering.

**[0016]** FIG. **6** illustrates the device positioned over a window as a drape.

**[0017]** FIG. **7** illustrates the device positioned over a window as a shade.

**[0018]** FIG. **8** illustrates the device hanging from a wall.

**[0019]** FIG. **9** illustrates the device hanging from a door.

**[0020]** FIG. **10** illustrates a controller for the device.

**[0021]** FIG. **11** illustrates an embodiment where the device is hanging and used as a shower curtain.

[0022] FIG. **12** illustrates an embodiment where the device is a room divider that used to modify environmental conditions in the room.

[0023] FIG. **13** is an exploded view of the multipurpose device of FIG. **1**.

## DETAILED DESCRIPTION

[0024] The present disclosure relates to a multipurpose device with controllable photonic modalities and/or electromagnetic modalities that can be utilized to provide controlled photonic energy or biometric energy to a user or to provide photonic and/or electromagnetic energy to affect the environmental conditions in a space. The device includes a substrate that can be rolled into a coil for storage or can be unrolled to provide an area where photonic and/or electromagnetic energy can be controllably emitted.

[0025] In some embodiments the device is configured to be used as a mat on a floor. In other embodiments, the device can be hung vertically from a wall, a door or a rod, such as a window curtain rod, a shower rod or a room divider rod.

[0026] By way of non-limiting example, the device can emit photonic energy in the ultraviolet spectrum (UVC range (100-280 nm), UVB range (280-315 nm) and UVC range 315-400 nm)) which can destroy microorganisms, remove noxious odors, cleansing and purifying the air, generate ozone and to reduce the presence of fungus, viruses, parasites and/or bacteria on the device. The device can emit photonic energy in the visible light spectrum (400-720 nm) or in any of the individual color spectrums, including but not limited to, colors in the red, green and blue spectrum or full spectrum white light, all of which can provide benefits for a user as well as enhance the ambiance of a space. Additionally, the device can be used to emit photonic energy in the infrared spectrum for wellness benefits and/or heating. The device can also optionally emit electromagnetism such as, but not limited to, pulsed electromagnetic field energies that can create geomagnetism fields and/or restore or enhance ambient geomagnetic and/or electromagnetic gauss levels.

[0027] The device can optionally include selected natural stones such as jade, tourmaline, bian stones, quartz, germanium or crystals; salt blocks, minerals, man-made materials, other natural materials such as, but not limited to, iron, nickel and cobalt, all of which emits photonic energy, natural geomagnetism, combinations thereof and/or that provide wellness benefits. The photonic energy and/or electromagnetic energy emitted from the natural and man-made materials can occur naturally and can be enhanced by heating or applying electric current the material. The electromagnetic energy is emitted the range of 1-100 Hz, where humans emit electromagnetic frequencies in the range of 9-16 Hz; average for males emit electromagnetic energy at about 12.2 Hz, average females emit electromagnetic energy at about 12.8 Hz for females and the mean human average is 12.3 Hz. The combination of electromagnetic energy and heat have been found to be useful in treating musculoskeletal issues, circulation issues, nervous system issues, digestion issues, respiration issues, wounds and other areas of concern. Typical ranges of Hz emitted by the device ranges from about 1 Hz to about 100 Hz and more typically, about 1 Hz to about 30 Hz and even more particularly, about 1 Hz to about 22 Hz.

[0028] Referring to FIG. **1**, a multipurpose device with controllable photonic modalities is illustrated at **10**. The multipurpose device **10** includes a substrate **12** having a top or front surface **14** onto which one or more arrays of controllable photonic emitters **20** is attached. The substrate **12** with the one or more arrays of controllable photonic emitters **20** is configured to be rolled into a coil **16** and retained in a sleeve **18** for compact storage and to be unrolled such that the substrate **12** and the array of controllable photonic emitters **20** do not overlap.

[0029] The array of controllable photonic emitters **20** includes emitters **22** that emit light in the visible spectrum and optionally the infrared spectrum. The emitters **22** are controllable to emit a selected wavelength corresponding to a color. While all colors in the visible spectrum can be emitted, the emitters **22** can be controlled to emit at least white light, red light, blue light and green light. The array of controllable emitters **20** can also include emitters **24** that emit photonic energy in the infrared spectrum and optionally emitters **26** that emit photonic energy in the ultraviolet spectrum

[0030] In some embodiments, the device **10** includes electromagnetic energy emitters, including but not limited to, one or more pulsed electromagnetic field motor **28**, that are configured to emit pulsed electromagnetic frequency energy. The pulsed electromagnetic frequency energy creates magnetic fields that restore or enhance magnetic flux levels that can be beneficial to a user's wellness. In some embodiments, the device **10** includes natural materials **29** that emit electromagnetic energy such as, but not limited to, iron, nickel and cobalt.

[0031] The device **10** can also optionally include natural stones **30** that emit photonic energy. In other embodiments, the device **10** can include salt mineral blocks **32** to trace minerals and provide halotherapy that ionizes and cleans the ambient air. In yet other embodiments, the device **10** can include man-made materials **34** that emit photonic and/or electromagnetic energy.

[0032] The inclusion of the infrared emitters **24**, the ultraviolet emitters **26**, the pulsed electromagnetic field motors **28**, the natural stones **30**, the salt blocks **32** and the man-made stones **34** are optional by themselves or in combination to form different applications. For instance, when the device **10** is used for a mat, the device can include visible light emitters **22** and infrared emitters **24**. When the device is hung from a wall or door, the infrared emitters **24**, the ultraviolet emitters **26**, the pulsed electromagnetic field motors **28**, the natural materials **29** that emit biomagnetic energy, the natural stones **30**, the salt mineral blocks **32** and the man-made stones **34** can be included as the user will not be in contact with the device **10**.

[0033] The device **10** includes a plurality of grommets **40** located about the perimeter **42**. The grommets **40** can be used to hang the device from a surface. Alternatively, the grommets **40** can used to retain a substantially optically transparent and substantially water impermeable plastic cover on the device to prevent sweat from permeating the substrate.

[0034] Referring to FIGS. **1** and **5**, the device **10** is configured for use as a mat on a floor surface. The substrate **12** for use as a mat includes closed cell foam or tightly woven cloth material. The substrate **12** of closed cell foam or tightly woven cloth material provides cushion and comfort to the user, whether the user is passively sitting or laying on the mat or actively exercising, performing yoga, meditating or receiving a massage. Additionally, the substrate **12** is durable and resistant to wear, which can extend the useful life of the device. Referring to FIG. **5**, a plastic film **42** that is substantially optically transparent and substantially water impermeable plastic cover can be placed onto the top surface **14** and retained with ties **44** to retain the plastic film **42** on the substrate **12**. An exemplary material of construction for the plastic film **42** is polyvinylchloride (PVC), however other plastic materials are also within the scope of the present disclosure. The plastic film **42** is used to prevent perspiration or other liquids from permeating the substrate **12** to maintain hygiene in the device **10**. In some instances, the film **42** is configured to be disposable with an intended one-time use.

**[0035]** Referring to FIGS. **1** and **13**, the device **10** includes a plurality of layers that are designed to cause electromagnetic energy, IR energy and visible light energy to be emitted through a top layer **99**. The device **10** includes a bottom layer **80** that is substantially moisture and wear resistant, so that the bottom layer prevents moisture from absorbing into the interior of the device and damaging the electronic equipment. The bottom layer **80** is wear resistant to prevent the bottom layer **80** from eroding due to wear with a surface as the user is engaging the top layer **9**.

**[0036]** The device includes an intermediate layer **82** that is positioned between the bottom layer **80** and a PEMF system layer **84** to which the electromagnetic field motors **28** are secured. The insulating layer **82** is substantially non-conductive to prevent electric energy from electromagnetic field motors **28** to conduct to ground. Rather, the PEMF energy from the electromagnetic field motors **28** is directed upwardly. An exemplary and non-limiting material of construction for the insulating layer **82** is a cotton sheet. However, other non-conductive are also within the scope of the present disclosure.

**[0037]** The device **10** includes a thermal insulating layer **84** that is positioned above the PEMF system layer **84**. The thermal insulating layer **84** is located below a thermal layer **86** that retains the IR emitters **22**. The thermal insulating layer **84** substantially prevents thermal energy from being directed toward the PEMF system layer **84** and protects the electromagnetic field motors **28** from the heat generated by the IR emitters **22**. As illustrated the IR emitters **22** includes a resistance heating wire. However, other IR emitters are within the scope of the present disclosure. An exemplary, non-limiting material of construction for the thermal insulating layer **84** includes a thermal foil insulation.

**[0038]** The layer **88** above the thermal layer **86** includes a sensor that senses the temperature created by the IR emitters **22** so that a controller can adjust the amount of current through the IR emitter **22** to control the temperature experienced by the user.

**[0039]** A layer **90** is configured to allow heat to raise from the thermal layer **86**, but has some insulating effects such that the user does not experience excessive heat.

**[0040]** A fireproof layer **92** is positioned above the layer **90** that prevents against user experiencing excessive temperatures. The fireproof layer **92** prevents the user from experiencing flames in the event a fire is caused below by the IR emitters **22** or a malfunction of the electromagnetic field motors **28**. An exemplary, non-limiting, material of construction of the fireproof layer **92** is a fireproof cotton layer.

**[0041]** An upper insulating layer **94** is above the fireproof layer **92**. The upper insulating layer **92** allows heat to gradually pass therethrough such that the temperature experienced by user engaging the top layer **99** gradually raises as the user is on the device **10**. An exemplary, non-limiting material of construction of the upper insulating layer **92** is a thermal foil insulation.

**[0042]** A waterproof layer **96** is above the upper insulating layer **94**. The waterproof layer **98** prevents moisture, such as sweat or spilled liquid, from affecting the insulating layers and/or the electronic equipment, including the IR emitters **22** and the electromagnetic field motors **28**. An exemplary, non-limiting material of construction of the waterproof layer **96** is polyvinylchloride polymer.

**[0043]** A layer **98** is above the waterproof layer **96**. The layer **98** is typically non-conductive and retains visible light emitters **22**. The visible light emitters **22** are typically light emitting diodes (LED) that are controllable between a plurality of colors of the spectrum and also white light, such that the user can experience the desired wavelengths. In other embodiments, the plurality of LEDS is configured to emit a selected wavelength.

**[0044]** The layer **97** is above the layer **98** and retains the natural stones **30** and optionally man-made stones **34**. The natural stones **30** are located above the IR emitters **22** such that the IR emitters **22** heat the natural stones **30** which causes the natural stones **30** to emit photons that are beneficial to the user.

**[0045]** The upper layer **99** is above the layer **97** and is constructed of a material that is comfortable to the touch of the skin to provide comfort to the user. The upper layer **99** also includes a transparent area that allows visible light to pass therethrough to provide phototherapy to the user. An exemplary material of construction of the upper layer **99** besides the transparent area is a polyurethane based material.

**[0046]** When used as a mat, the array of controllable photonic emitters **20** includes emitters **22** that emit light in the visible spectrum and optionally the infrared spectrum. Other prior mentioned photonic modalities including the ultraviolet emitters **26**, the pulsed electromagnetic field motors **28**, the natural materials **29** that emit electromagnetic energy, the natural stones **30**, the salt mineral blocks **32** and the man-made materials **34** can optionally be included in the **10** and supported by the substrate **12** to provide additional photonic and electromagnetic benefits and effects. When used as a mat, the device **10** typically has a length of at least about 5 feet and a width of at least about 2.5 feet. However, other length and width dimensions of the device are within the scope of the present disclosure.

**[0047]** Referring to FIGS. **1** and **13**, an exemplary, non-limiting construction of the device **10** includes a plurality of layers that are designed to cause electromagnetic energy, IR energy and visible light energy to be emitted through a top layer **99**. The device **10** includes a bottom layer **80** that is substantially moisture and wear resistant, so that the bottom layer prevents moisture from absorbing into the interior of the device and damaging the electronic equipment. The bottom layer **80** is wear resistant to prevent the bottom layer **80** from eroding due to wear with a surface as the user is engaging the top layer **9**.

**[0048]** The device includes an intermediate layer **82** that is positioned between the bottom layer **80** and a PEMF system layer **84** to which the electromagnetic field motors **28** are secured. The insulating layer **82** is substantially non-conductive to prevent electric energy from electromagnetic field motors **28** to conduct to ground. Rather, the PEMF energy from the electromagnetic field motors **28** is directed upwardly. An exemplary and non-limiting material of construction for the insulating layer **82** is a cotton sheet. However, other non-conductive are also within the scope of the present disclosure.

**[0049]** The device **10** includes a thermal insulating layer **84** that is positioned above the PEMF system layer **84**. The thermal insulating layer **84** is located below a thermal layer **86** that retains the IR emitters **22**. The thermal insulating layer **84** substantially prevents thermal energy from being directed toward the PEMF system layer **84** and protects the electromagnetic field motors **28** from the heat generated by the IR emitters **22**. As illustrated the IR emitters **22** includes a resistance heating wire. However, other IR emitters are within the scope of the present disclosure. An exemplary, non-limiting material of construction for the thermal insulating layer **84** includes a thermal foil insulation.

**[0050]** The layer **88** above the thermal layer **86** includes a sensor that senses the temperature created by the IR emitters **22** so that a controller can adjust the amount of current through the IR emitter **22** to control the temperature experienced by the user.

**[0051]** A layer **90** is configured to allow heat to raise from the thermal layer **86**, but has some insulating effects such that the user does not experience excessive heat.

**[0052]** A fireproof layer **92** is positioned above the layer **90** that prevents against user experiencing excessive temperatures. The fireproof layer **92** prevents the user from experiencing flames in the event a fire is caused below by the IR emitters **22** or a malfunction of the electromagnetic field motors **28**. An exemplary, non-limiting, material of construction of the fireproof layer **92** is a fireproof cotton layer.

**[0053]** An upper insulating layer **94** is above the fireproof layer **92**. The upper insulating layer **92** allows heat to gradually pass therethrough such that the temperature experienced by user engaging the top layer **99** gradually raises as the user is on the device **10**. An exemplary, non-limiting material of construction of the upper insulating layer **92** is a thermal foil insulation.

**[0054]** A waterproof layer **96** is above the upper insulating layer **94**. The waterproof layer **98** prevents moisture, such as sweat or spilled liquid, from affecting the insulating layers and/or the electronic equipment, including the IR emitters **22** and the electromagnetic field motors **28**. An exemplary, non-limiting material of construction of the waterproof layer **96** is polyvinylchloride polymer.

**[0055]** A layer **98** is above the waterproof layer **96**. The layer **98** is typically non-conductive and retains visible light emitters **22**. The visible light emitters **22** are typically light emitting diodes (LED) that are controllable between a plurality of colors of the spectrum and also white light, such that the user can experience the desired wavelengths. In other embodiments, the plurality of LEDS is configured to emit a selected wavelength.

**[0056]** The layer **97** is above the layer **98** and retains the natural stones **30** and optionally man-made stones **34**. The natural stones **30** are located above the IR emitters **22** such that the IR emitters **22** heat the natural stones **30** which causes the natural stones **30** to emit photons that are beneficial to the user.

**[0057]** The upper layer **99** is above the layer **97** and is constructed of a material that is comfortable to the touch of the skin to provide comfort to the user. The upper layer **99** also includes a transparent area that allows visible light to pass therethrough to provide phototherapy to the user. An exemplary material of construction of the upper layer **99** besides the transparent area is a polyurethane based material.

**[0058]** Once the stack of layers is constructed, the edges of at least the lower layer **80** and the upper layer **99** are secured together to retain the stack of layers in alignment. In some embodiments, the edges are sewn together and in other embodiments that edges are bonded together.

**[0059]** Referring to FIGS. **2-4**, the device **10** is used as cover or sheet for the person to sit or lay upon, where the area of the device **10** is determined based upon the required area needed to provide the treatment. Referring to FIG. **2**, the device **10** is used as a cover for a lounging chair **50**. When the user reclines on the lounging chair **50** the device **10** emits controlled photonic energy from the array of photonic emitters **20**, the natural stones **30** and/or the man-made stones **34**. The device can additionally and/or alternatively provide electromagnetic energy from the pulsed electromagnetic field motors **28** and/or the natural materials **29** that emit electromagnetic energy. Similarly, referring to FIG. **3**, the device **10** can be draped over a couch **52** or other sitting furniture or appliance where the user sits on the device **10** to receive controlled photonic energy and/or electromagnetic energy. Referring to FIG. **4**, the device **10** can also be used as a sheet or spread on a bed **54**.

**[0060]** The device **10** can also be hung in numerous locations within a location to provide a person with the controlled photonic energy and/or electromagnetic energy, or to change the environmental conditions and/or to provide ambiance. Referring to FIG. **6**, the device **10** can be hung over a window **56** with a rod **57** to provide the controlled photonic energy and/or electromagnetic energy while functioning as a curtain. Referring to FIG. **7**, the device **10** can be coiled about a spring biased roller **58** to function as a shade or blind over a window **59**, where the device can be extended to allow the person with the controlled photonic energy and/or electromagnetic energy, or to change the environmental conditions and/or to provide ambiance and to be retracted to allow natural light and/or geomagnetism into the space.

**[0061]** Referring to FIG. **8**, the device **10** can be hung from a wall **60** using loops or carabiners **62** positioned through the grommets **40** where the carabiners or loops **62** attached to hooks **64** extending from the wall **60**. Referring to FIG. **9**, the device **10** is illustrated being attached to a door **70**. Bottom hook portions **72** of a retaining member **74** are positioned through the grommets **40** and top hook portions **76** extend over the top of the door **70** to retain the device **10** in an elevated position on the door **70**. Optional straps **78** secured to the side edges of the device **10** can be used to retain the device **10** to the door **70** as the door **70** is opened and shut.

**[0062]** The device **10** can be used to provide controlled photonic energy and/or electromagnetic energy to a user by having the user stand proximate the device **10**, as illustrated in FIGS. **6-9**. Alternatively, the device can be used to control and or modify the ambient conditions in a space, including but not limited to emitting visible light to change the lighting, emitting infrared energy to provide heat, emitting ultraviolet energy to destroy microorganisms, remove noxious odors, cleanse and purify the air and/or generate ozone. As the device **10** is used as a hanging device with little to no contact with the user, the substrate **12** can be less robust than a device used by the user for sitting or reclining. The hanging device **10** can utilize a less heavy tightly woven fabric or a sheet of cloth or plastic material to support the elements emitting photonic energy, electromagnetic energy and/or halotherapy.

**[0063]** The size of the hanging device **10** can be varied depending upon its use. When hung from the door **70**, the area of the device **10** should be less that the area of the door **70**. However, when hung from a wall **60**, the size of the device can vary from the size used as a mat to provide photonic energy, electromagnetic energy and/or halotherapy to a user to the area of the wall when used to modify ambient conditions in the space.

**[0064]** Additionally, the device can include one or more of the mentioned photonic modalities and/or the mentioned electromagnetic modalities by themselves or in combination to provide the desired treatment for the user or modify the ambient conditions in the space. The photonic modalities for use in the hanging device **10** include the controllable visible light emitters **22**, the infrared emitters **24**, the ultraviolet emitters **26**, the natural stones **30**, the salt blocks **32** and the man-made stones **34** by themselves or in any desired combination while the biomagnetic modalities include the pulsed electromagnetic field motors **28** and/or the natural materials **29**.

**[0065]** The device **10** is powered by a standard 110 volt power supply and includes a cord **11** that has a plug that plugs into a standard electrical outlet. The cord **11** is sufficiently long to provide a desired range of movement to place or hang the device **10** in a selected location.

**[0066]** The device **10** includes a corded or battery-operated controller **100** as illustrated in FIGS. **1** and **10**. The controller **100** includes an on/off button **102** that allows the user to operate the device **10**. The controller **100** includes a timer function **102** that allows the duration of use to be set. A shift button **104** allows the user to select one of a plurality of preset times, including for instance, 10 minute intervals starting at 10 minutes and ending at 60 minutes. A screen **106** allows the user to visual the control adjustments being made. The button **108** allows for the selection of wavelengths in the visible spectrum, including but not limited to, red, green, blue and white. The selection is made using the shift button **104** and viewing the screen **106**. Typically, at least near infrared photonic energy is emitted with the selected visible photonic energy. Once the wavelength is selected, the intensity or brightness button **110** is engaged and the shift button **104** is used to provide the desired intensity. The visible light can be pulsed by selecting the button **112** and utilizing the shift button **104** and screen **106** to control the pulse rate. By way of non-limiting example, the pulse rate can be controlled between 1 and 999 Hz.

**[0067]** The button **114** can be utilized to energize the pulsed electromagnetic field motors **28**, where the shift button **104** and screen can be utilized to set the rate and intensity of the electromagnetic pulses. The button **116** can be utilized to energize the ultraviolet emitters **26** where the shift button **104** and the screen **106** are utilized to control the ultraviolet emitters. Other control features can also be added to the controller as needed to provide the selected photonic and/or electromagnetic energy for a selected purpose.

**[0068]** Additionally, the controller **100** can include preset protocols for specific purposes. Table 1 below provides a sample of present protocols that can be utilized.

TABLE 1

| Desired Effect | Time (Min) | Red | Green | Blue | White | NIR | EM (Hz) |
|---|---|---|---|---|---|---|---|
| Cellulite Smoothing | 15 | Y | N | N | Y | Y | Y |
| Skin Toning | 10 | Y | Y | N | Y | Y | Y |
| Body Cleansing | 10 | Y | Y | Y | N | Y | Y |
| Weight Management | 15 | Y | N | Y | N | Y | Y (8-12) |
| Strength and Recovery | 15 | Y | N | N | Y | Y | Y (12) |
| Muscle Relaxation | 30 | Y | N | N | N | Y | Y (10) |
| Energy and Endurance | 30 | N | N | Y | N | Y | Y (12) |
| Relaxation and Stress Relief | 30 | Y | N | N | N | Y | Y (20) |
| Gentle Cleanse | 30 | N | Y | N | N | Y | Y (7) |
| Flexibility | 15 | Y | N | N | Y | Y | Y (16-20) |
| Toxic Deep Cleanse | 15 | Y | Y | N | N | Y | Y (10) |
| Calm & Clarity | 15 | N | Y | N | Y | Y | Y (14-20) |
| Mood Boost | 15 | N | N | Y | Y | Y | Y |
| Mind Wellness (PM) | 30 | N | N | N | Y | Y | Y |
| Mind Wellness (AM) | 30 | N | N | Y | Y | Y | Y |
| Deep Sleep (PM) | 30 | Y | N | N | N | Y | Y (5) |
| Deep Sleep (AM) | 30 | Y | N | N | Y | Y | Y (5) |

**[0069]** The preset protocols allow the user to efficiently receive the desired controlled photonic energy without having to adjust the protocol for curtained desired effects. It is understood the protocols can include electromagnetic sessions in desired Hertz and time periods along with the photonic treatments or by itself.

**[0070]** Referring to FIG. **11**, another embodiment of the device is illustrated at **200**. The device **200** is used as a shower curtain that is supported by a shower rod **202** that spans the length of a bath tub **204** or shower area. Loops **206** are positioned through grommets and about the shower rod **202** to suspend the shower curtain **200**. The shower curtain is sized to span a length of the bath tub **204** or shower area and has a height where a bottom edge **201** extends below the top of the bath tub **204** or shower area.

**[0071]** The device **200** includes a back substantially water impermeable sheet **210** that can be optically reflective and a front substantially water impermeable sheet **212** that is substantially optically transparent. Arrays of LED light emitters and optional near infrared (NIR) emitters **214** are positioned between the two sheets **210** and **212** and the LED and/or NIR lights **214** are sealed between the two sheets **210** and **212**. A cord **216** extends from a top corner of the shower curtain away from a shower head **205** to aid in prevent moisture from engaging the cord **216**. The cord **216** is in electrical communication with a control panel **218** that is powered by a battery pack **220** that can have a unique interface with a receptacle **222**.

**[0072]** The power in the battery pack is calculated by determining the power needed to power the control panel **218** and the array LED light emitters and/or NIR emitters **214** for a selected period of time. The power in the battery pack is sufficiently low so that no harm would come to a user in the event of a failure of the seal between the sheets **210** and **212**.

**[0073]** The control panel **218** allows the user to select the wavelength, intensity and pulsation frequency of the visible light and/or NIR as well as the duration of use. In some embodiments, the device **200** includes the pulsed electromagnetic field motors **205** and/or the natural materials **207** that emit biomagnetic energy. The user can then bathe or shower while received a controlled phototherapy session and/or a biomagnetic session.

**[0074]** Referring to FIG. **12**, another embodiment of the device is illustrated at **300** where the device **300** is used as room divider. The device **300** is positionable about a bed **302** where the device is able to change ambient conditions for a person using the bed **302**. The device **300** can be utilized to provide photic energy and/or electromagnetic energy for treatment. The device **300** can also include larger densities of UV emitters such that the devices can be used to sanitize and disinfect the atmosphere in a room, such as a multi-person hospital room, where the UV light may be advantageous is prevent a person from become ill with airborne bacteria, fungi and/or viruses.

**[0075]** Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above as has been determined by the courts. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

**1**. A device for providing controlled photonic modalities and electromagnetic modalities, the device comprising:
- a substrate having a length and a width;
- at least one array of controllable photonic emitters attached the substrate;
- at least one electromagnetic energy emitter attached to the substrate; and
- a controller configured to allow a user to select the type of photonic energy and the electromagnetic energy.

**2**. The device of claim **1**, wherein the substrate comprises a tightly woven cloth material or a closed cell foam material.

**3**. The device of claim **1**, wherein the substrate comprises a sheet of cloth material.

**4**. The device of claim **1** wherein the substrate comprises first and second substantially water impermeable sheets and wherein the at least one array of controllable photonic emitter are sealed between the first and second sheets.

**5**. The device of claim **4**, wherein the first sheet is substantially optically reflective and the second sheet is substantially optically transparent.

**6**. The device of claim **1**, wherein the at least one array of controllable photonic emitters comprises an array of controllable LED emitters and/or infrared emitters.

**7**. The device of claim **1** wherein the substrate is further configured to support one or more of emitters that emit photonic energy in infrared spectrum, emitters that emit photonic energy in the ultraviolet spectrum, one or more pulsed electromagnetic field motor that are configured to emit pulsed electromagnetic frequency energy, natural materials that emit electromagnetic energy, natural stones, salt mineral blocks and man-made materials.

**8**. The device of claim **1**, wherein the device is configured to be used as a mat, a sheet or a cover supported on a surface wherein a user physically engages the device.

**9**. The device of claim **1**, wherein the device is configured to be hung from a door, a wall or a window.

**10**. The device of claim **9**, wherein the hung device is configured to be utilized to provide photonic energy and/or electromagnetic energy to a user.

**11**. The device of claim **9**, wherein the hung device is configured to enhance and/or to modify ambient conditions in a space.

**12**. The device of claim **4**, wherein the device is configured for use as a shower curtain, a room divider and/or an ambient modifier of a space.

**13**. The device of claim **12**, wherein the device is powered by a battery pack.

**14**. The device of claim **13**, wherein the control is preprogrammed with a plurality of protocols.

**15**. The device of claim **1**, wherein the controller is configured to adjust a duration and/or a pulse rate of the photonic energy and/or electromagnetic energy.

**16**. A device for providing controlled photonic modalities and electromagnetic modalities, the device comprising:
- a base layer having an upper surface and a lower surface configured to engage a surface;
- a first internal layer supporting pulsed electromagnetic field motor, the first internal layer having a lower surface facing the upper surface of the base layer and an upper surface;
- a second internal layer supporting one or more infrared emitters, wherein the second internal layer having a lower surface facing the first internal layer and an upper surface;
- a first internal insulating layer between the first internal layer and the second internal layer;
- a third internal layer supporting a plurality of visible light emitters, the third internal layer having a lower surface facing the upper surface of the second internal layer
- a second internal insulating layer between the second internal layer and the third internal layer;
- a fourth internal layer supporting a plurality of stone that emit photons when heated, the fourth internal layer having a lower surface facing the upper surface of the third internal layer and an upper surface; and
- an upper layer having a substantially optically transparent area, the upper layer having a lower surface facing the upper surface of the fourth internal layer and an upper surface configured to engaged by a user.

**17**. The device of claim **16** and further comprising:
- a substantially non-conductive layer positioned between the base layer and first internal layer.

**18**. The device of claim **16** and further comprising:
- a temperature sensor positioned the second internal layer and the second internal insulating layer.

**19**. The device of claim **16** and further comprising:
- a substantially waterproof layer positioned between the second internal insulating layer and the third internal layer.

**20**. The device of claim **16** and further comprising:
a substantially fireproof layer positioned between the second internal layer and the second internal insulating layer.

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